

Long-term recordings dataset, 2018

Long term recordings have been performed in an incubator at 5% CO₂, 37°C while all MEAs had been sealed with PDMS caps. Extracellular raw electrical signals were sampled at 10 kHz/channel and each recording session had been split on 30 min intervals of continuous recording. We performed 4 types of experiments (4 folders) and for each experiment 6 distinct MEAs had been used (6 subfolders).

List of folders	6 subfolders with one individual MEA	
Cntr	24 h long term recordings of spontaneous activity	Each subfolder contains 48 files of the 30 min recording
GBZ	8 h control recordings 8 h application 30 μ M, Gabazine, GBZ	Each subfolder contains 16 files of recording before and 16 files after drug application
AP5	8 h control recordings 8 h application 15 μ M, D-2-Amino-5-phosphonopentanoic acid, AP5	
CTZ	8 h control recordings 8 h application 10 μ M, cyclothiazide, CTZ	

The filename of recordings identifies culture ID, condition of recordings (control /drug application), MEA ID, order of recordings:

420_cn_I15918_0004.mat

420: culture identifier

cn: control activity recordings

I15918: MEA name

0004: order of recordings

Each file contains a double array **spiketime**, where the first column is the spikes time (from 0 to 1800000 msec), the second is the #electrodes which detected a spikes (from 1 to 60).